18 PROBABILITY

Probability

The term probability refers to the chance of happening or not happening of an event.

Sample space

The set of all possible and distinct outcomes of an experiment is called the Sample Space of that experiment and is denoted by S. Each element of the sample space is called a sample point. n(s) denotes the number of elements of S.

Exhaustive cases

All possible out comes of an event are known as exhaustive cases.

Favorable cases

The number of outcomes which result in the happening of a desired event are called favorable cases.

Thus in a single throw of a dice the no. of favorable cases of getting an odd no. are three i.e. 1, 3 and 5.

Mutually exclusive events

Two or more events are said to be mutually exclusive if the happening of any one of them excludes the happening of all other in a single experiment. Thus in the throw of a single dice the event 5 and 6 are mutually exclusive .

Equally likely events

Two or more events are said to be equally likely if the chance of their happening is equal. In the throw of an unbiased coin the coming up of head or tail is equally likely.

Independent and dependent event

An event is said to be independent if its happening is not affected by the happening of other events. Thus in the throw of a dice repeatedly, coming up of 5 on the first throw is independent of coming up of 5 again in the second throw.

An event is said to be dependent if its happening is affected by the happening of other event. Thus if we are successively drawing cards from a pack the events would be dependent.

Formula to compute probability of an event

If an event can happen in 'm' ways and fail to happen in 'n' ways then the probability (P) of the happening of the event is given by

$$P = \frac{m}{m+n}$$

$$= \frac{\text{No. of cases favourable to the occurence of the event}}{\text{Total no. of exhaustive cases}}$$

Results on probability

(i) For any event A and B, we have :

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

(ii) If the events are mutually exclusive and exhaustive. The sum of the individual probabilities = 1

$P(a) + P(\overline{A}) = 1.$

So, $P(\overline{A}) = 1 - P(a)$

(iii) Range of probability is $0 \le P(E) \le 1$

Ex.1 Find the probability of getting at least one head when two coins are tossed.

⇒ S = [HH, TH, HT, TT] ∴ n(S) = 4

E = {HH, TH, HT} ∴ n[E] = 3

P(E) = ¾

Ex.2 Find the probability of getting a multiple of 2 in the throw of a die.

⇒ S = {1, 2, 3, 4, 5, 6}

∴ n(S) = 6

Let A be the event that the die shows a multiple of 2.

A = {2, 4, 6}

∴ n(A) = 3

$$P(A) = \frac{n(A)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

Ex.3 Find the probability that the birthday of a child is on a Saturday or Sunday.

⇒ n(a) = 2; n(S) = 7 then p(a) = 2/7

Ex.4 A bag contains 7 red, 5 blue, 4 white and 4 black balls. Find the probability that a ball drawn at random is red or white.

⇒ A ball can be selected from 20 in ${}^{20}C_1$ = 20 ways.

n(S) = 20

Let A be an event that the ball drawn is red or white.

∴ $n(A) = {}^{7}C_1 + {}^{4}C_1 = 7 + 4 = 11$

∴ p(a) = 11/20

Ex.5 A committee of 5 students is to be chosen from 6 boys and 4 girls. Find the probability that the committee contains exactly 2 girls.

⇒ 5 students can be selected from 10 in ${}^{10}C_5$ ways

n(S) = 252

n(a) = ${}^{4}C_2 \times {}^{6}C_3 = 6 \times 20 = 120$

p(a) = $\frac{n(A)}{n(S)} = \frac{120}{252} = \frac{10}{21}$

EXERCISE

1. A and B are two mutually exclusive events such that P (a) = $\frac{1}{3}$ and P (b) = $\frac{1}{5}$. Find $P(A \cup B)$

(a) $\frac{1}{15}$ (b) $\frac{8}{15}$

(c) $\frac{7}{15}$ (d) None of these

2. From a pack of 52 cards, two are drawn one by one without replacement. Find the probability that both of them are jacks.

(a) $\frac{1}{221}$ (b) $\frac{11}{850}$ (c) $\frac{25}{204}$ (d) $\frac{5}{24}$

3. 4 cards are drawn from a well-shuffled pack of 52 cards. Find the probability that they are of black color or face cards.

(a) $\frac{2}{5}$ (b) $\frac{3}{5}$ (c) $\frac{4}{5}$ (d) None

4. The probability of raining on day 1 is 0.3 and on day 2 is 0.5, What is the probability of raining on both the days ?

(a) 0.2 (b) 0.15 (c) 0.06 (d) 0.25

5. If a dice is thrown 4 times, what is the probability that an odd number will come up exactly 3 times ?

(a) $\frac{8}{16}$ (b) $\frac{1}{4}$ (c) $\frac{3}{16}$ (d) $\frac{5}{16}$

6. What is the probability that there are 53 Mondays and 53 Tuesdays in a leap year ?

(a) 1/7 (b) 2/7

(c) 0 (d) None of these

7. Two dice are thrown. The probability that the same number will appear on each of them is

(a) $\frac{1}{6}$ (b) $\frac{1}{18}$

(c) $\frac{1}{36}$ (d) None of these

8. A bag contains 7 green and 3 blue balls. Two balls are drawn at random. Find the probability that they are of the same color.

(a) $\frac{7}{15}$ (b) $\frac{3}{5}$

(c) $\frac{8}{15}$ (d) None of these

9. 3 coins are tossed simultaneously, what is the probability of getting at least one tail ?

(a) $\frac{1}{8}$ (b) $\frac{3}{8}$ (c) $\frac{7}{8}$ (d) 1

10. If the probability of A failing in an competition is $\frac{4}{5}$ and that of B failing the same competition is $\frac{7}{10}$, then the probability that either A or B fails is :

(a) $\frac{1}{2}$ (b) $\frac{11}{25}$

(c) $\frac{19}{50}$ (d) None of these

11. 51 tickets are numbered from 1 to 51. One ticket is drawn at random out of these tickets. What is the probability that the ticket number is a multiple of 2 or 5 ?

(a) $\frac{2}{5}$ (b) $\frac{1}{5}$

(c) $\frac{3}{5}$ (d) None of these

12. Out of 13 candidates for a job, there are 4 girls and 9 boys. It is desired to select 2 persons for the job. The probability that atleast one of the selected person will be a girl is :

(a) 25/39 (b) 5/13 (c) 14/39 (d) 7/13

13. Pankaj has 10 pairs of dark black socks and 10 pairs of red socks. He keeps them all in the same bag. If he picks out three socks at random, then what is the probability that he will get a matching pair ?

(a) $(2^{10}C_2{}^{10}C_1)/^{20}C_3$ (b) $(^{10}C_3{}^{10}C_1)/^{20}C_3$

(c) 1 (d) None of these

14. A bag contains 5 blue, 3 black and 10 red balls. If three balls are drawn at random, find the probability that one is blue, one is black and one is red.

(a) $\frac{4}{21}$ (b) $\frac{5}{147}$ (c) $\frac{25}{136}$ (d) $\frac{9}{19}$

15. In a race where 3 horses are running, the chance that horse A will win is $\frac{1}{5}$, that B will win is $\frac{1}{15}$ and that C will win is $\frac{1}{10}$. Assuming that a dead heat is impossible. Find the chance that one of them will win.

(a) $\frac{11}{30}$ (b) $\frac{1}{480}$ (c) $\frac{1}{160}$ (d) $\frac{1}{240}$

16. In an essay competition, the odds in favor of competitors A, B, C, D are 1 : 4, 1 : 5, and 1 : 6 and 1 : 7 respectively. Find the probability that one of them wins the competition.

(a) $\frac{114}{120}$ (b) $\frac{1}{120}$ (c) $\frac{319}{420}$ (d) $\frac{17}{114}$

17. In a bag containing 13 white and 5 black balls. Two balls are chosen at random and the first one is found to be white. The probability that the second one is also white is.

(a) $\frac{11}{17}$ (b) $\frac{12}{17}$ (c) $\frac{13}{18}$ (d) $\frac{5}{17}$

18. If the probability of A to fail in an examination is 0.2 and that of B is 0.3, then the probability that either A or B fails is

(a) 0.5 (b) 0.06 (c) 0.44 (d) 0.1

19. In a class of 11 pupils, 5 are boys and the remaining are girls. From it 2 pupils are selected at random. The probability that both are girls, will be.

(a) $\frac{5}{11}$ (b) $\frac{6}{11}$ (c) $\left(\frac{6}{11}\right)^2$ (d) $\frac{3}{11}$

20. The probability that a company executive will travel by plane is $\frac{2}{5}$, and will travel by train is $\frac{1}{3}$. Find the probability of his travelling by plane or by train.

(a) $\frac{11}{15}$ (b) $\frac{2}{15}$ (c) $\frac{3}{5}$ (d) $\frac{1}{15}$

21. What is the chance of throwing a number greater than 3 with an ordinary dice whose faces are numbered from 1 to 6 ?

(a) $\frac{2}{3}$ (b) $\frac{2}{5}$ (c) $\frac{1}{3}$ (d) $\frac{1}{2}$

22. Two balls are to be drawn from a bag containing 4 red and 6 yellow balls, find the chance that both balls will be yellow.

(a) $\frac{7}{12}$ (b) $\frac{5}{3}$ (c) $\frac{1}{3}$ (d) $\frac{10}{22}$

23. Out of 10 outstanding students in a college, there are 4 boys and 6 girls. A team of five students is to be selected for a program. Find the probability that three are boys and two are girls.

(a) $\frac{11}{25}$ (b) $\frac{9}{11}$ (c) $\frac{10}{21}$ (d) $\frac{5}{21}$

24. A bag contains 21 tickets numbered from 1 to 21. A ticket is drawn and then another ticket is drawn without replacement. Find the probability that both tickets will show even numbers.

(a) $\frac{5}{12}$ (b) $\frac{3}{14}$ (c) $\frac{4}{19}$ (d) $\frac{9}{11}$

25. What is the probability that a number selected from the numbers 1, 2, 3,….,30 is prime number, when each of the given numbers is equally likely to be selected ?

(a) $\frac{3}{25}$ (b) $\frac{1}{3}$

(c) $\frac{11}{30}$ (d) None of these

26. One card is drawn from a pack of 52 cards, each of the 52 cards being equally likely to be drawn. Find the probability that the card drawn is red and a queen.

(a) $\frac{1}{13}$ (b) $\frac{7}{13}$ (c) $\frac{1}{26}$ (d) $\frac{1}{2}$

27. A word consists of 10 letters; 6 consonants and 4 vowels. Three letters are chosen at random. What is the probability that more than one vowel will be selected ?

(a) $\frac{17}{42}$ (b) $\frac{1}{3}$ (c) $\frac{1}{7}$ (d) $\frac{4}{5}$

28. A bag containing 4 black and 8 red, balls, two balls are drawn from the bag in succession without replacement. What is the probability that both the balls are red ?

(a) $\frac{1}{7}$ (b) $\frac{5}{11}$ (c) $\frac{13}{33}$ (d) $\frac{14}{33}$

29. The probability of success of three students P, Q and R in one examination are $\frac{1}{2}, \frac{1}{3}$ *and* $\frac{1}{4}$ respectively. Find the probability of success of at least two.

(a) $\frac{1}{6}$ (b) $\frac{7}{25}$ (c) $\frac{7}{24}$ (d) $\frac{3}{5}$

30. A boy and his friend appear in an interview for two vacancies in the same post. The probability of boy's selection is (1/3) and the probability of his friend's selection is (1/5). What is the probability that only one of them is selected ?

(a) $\frac{4}{5}$ (b) $\frac{2}{7}$ (c) $\frac{8}{15}$ (d) $\frac{2}{5}$

31. Find the probability that the number chosen randomly from the first 1000 natural numbers is a multiple of 4 or a perfect square.

(a) $\frac{530}{200}$ (b) $\frac{133}{500}$ (c) $\frac{381}{1000}$ (d) $\frac{1093}{4000}$

32. The probability that a man will get a plumbing contract is $\frac{2}{3}$.The probability that he will not get an electric contract is $\frac{5}{9}$. If the probability of getting at least one contract is $\frac{4}{5}$, what is the probability that he will get both the contracts?

(a) $\frac{14}{9}$ (b) $\frac{24}{45}$ (c) $\frac{14}{45}$ (d) $\frac{7}{9}$

33. There are 30 coins out of which 20 are biased. Out of the 20 biased coins, 15 always show head while the rest always show tails. The probability of getting at least 14 tails in a single throw of 30 coins is:

(a) 10×2^{-40} (b) 11×2^{-20}
(c) 10×2^{-10} (d) 11×2^{-10}

34. From a pack of 52 well shuffled cards, if two cards are drawn one after the other without replacement, what is the probability that neither of them is a 5 or a 7?

(a) $\frac{473}{668}$ (b) $\frac{4}{51}$ (c) $\frac{473}{663}$ (d) None

35. If $\frac{1-P}{3}, \frac{4P-1}{2}, \frac{6P-1}{4}$ are the probabilities of three mutually exclusive events, then p lies in which of the following intervals?

(a) $\left[\frac{1}{14}, \frac{3}{14}\right]$ (b) $\left[\frac{1}{4}, \frac{17}{38}\right]$ (c) $\left[\frac{1}{6}, \frac{17}{38}\right]$ (d) $\left[\frac{5}{38}, \frac{17}{38}\right]$

36. Each cell in a 4 × 2 grid is to be painted with one colour out of four colours red, blue, green and yellow, such that no two squares painted with the same colour are in the same row or in the same column. Find the number of ways to do this.

(a) 4! × 8! (b) 23 × 4! (c) 9 × 2! (d) 9 × 4!

37. Raju is a movie freak. Last Sunday, he watched five movie shows one after the other in his favourite multiplex. This multiplex has six screens, each screen showing a different movie. On each screen, a movie is repeated five times a day. If Raju watched exactly one movie twice and the other three movies only once, then in how many ways could he have watched the movies?

(a) $6 \times \frac{3!}{2!}$ (b) $60 \times \frac{6!}{2!}$
(c) $6 \times \frac{6!}{2!}$ (d) $6 \times \frac{5!}{2!}$

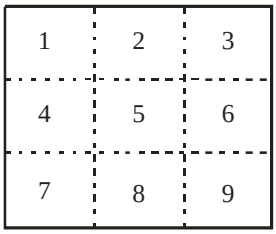
38. Amelia, Betty, Cindy, Dorothy, Elly and Frida are seated on chairs numbered 1 to 6 respectively. Their birthdays fall on the 21st, 2nd, 19th, 31st, 18th and 5th, respectively of the same month. If the seat number of a lady matches with the remainder of her birth date when the birth date is divided by 15, we say that she is "properly seated" else we say that she is "wrongly seated". How many of these ladies are 'wrongly seated' and in how many other ways could these wrongly seated ladies have been 'wrongly seated'?

(a) 5, 9 (b) 5, 44 (c) 5, 74 (d) 5, 43

39. There are 3 books on Mathematics, 4 books on Geography and 6 books on Microbiology (all these books are distinguishable) which are to be arranged on a shelf. Let A be the event that no two books on Microbiology are together and B be the event that all the books on the same subject are always together. Find the ratio of the probability of the event A occurring to that of the probability of event B.

(a) $\frac{490}{3}$ (b) $\frac{30}{7}$
(c) $\frac{75}{221}$ (d) $\frac{81}{65}$

Directions (40 - 41) : Answer the questions on the basis of the data given below.



The figure given above shows a square sheet of paper which, when folded along the dotted lines, is divided into nine congruent squares. An open box (cube) is to be formed using this sheet of paper such that:

(i) Each face of the cube to be formed is one of the small squares.

(ii) The square that form the cube must not be cut out separately; instead they must be folded along the dotted lines to form the cube.

(iii) There should be no overlap o f any two squares.
The squares that are not required to form the cube may be cut out from the sheet of paper.

40. If only square 1 is colored yellow, then in how many ways can a cube with one yellow face be formed?

(a) 6 (b) 2 (c) 11 (d) 8

41. If the squares in first row are colored red, those in the second row are colored green and those in the third row are colored blue, then in how many ways can we form a cube whose faces are of only 2 colours?

(a) 6 (b) 5 (c) 8 (d) None

42. Amit and Bhargav are playing a game of cards. They play n games in all. Each of them gets 1 points for their first win, 2 points for their respective second wins, 4 points for their respective third wins and so on. Amit wins more games than Bhargav does. The sum of ht epoints earned by Amit and Bhargav is 574. How many games does Amit win?

(a) 9 (b) 7 (c) 5 (d) 6

43. The odds are 7 to 5 against A, a person who is now 30 years old for living up to 70 yrs and the odds are 2 to 3 in favour of B who is now 40 yrs for living up to 80 yrs. What is the probability that at least one of these two persons will be alive 40 yrs hence?

(a) $\frac{18}{20}$ (b) $\frac{13}{14}$ (c) $\frac{13}{20}$ (d) $\frac{11}{12}$

44. In a residential colony, 60% are males. If 5% of the males are children (boys) and 10% of the females are children (girls), find the probability that a child selected at random happens to be a boy.

(a) $\frac{5}{14}$ (b) $\frac{3}{7}$ (c) $\frac{108}{203}$ (d) $\frac{4}{7}$

45. The probabilities of two events are such that

$P(A \cup B) \geq \frac{2}{5}, \frac{1}{10} \leq P(A \cap B) \leq \frac{2}{10}$

If on a graph, P(A) is represented on the X-axis and P(b) is represented on the Y – axis, then what is the area of the region (on the graph) in which every point satisfies the above relations ?

(a) $\frac{17}{20}$ (b) $\frac{119}{200}$

(c) $\frac{19}{40}$ (d) $\frac{40}{70}$

NOTES :

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1.(b) P (A $\cup$ B) = P(a) + P(b)

$\Rightarrow \frac{1}{3}+\frac{1}{5}=\frac{5+3}{15}=\frac{8}{15}$

2.(a) Required probability

$= \frac{4}{52}\times\frac{3}{51}=\frac{1}{13}\times\frac{1}{17}=\frac{1}{221}$

3.(d) Required probability

$= \frac{{}^{26}C_4+{}^{12}C_4}{{}^{52}C_4}$

4.(b) Required probability

= P(a) × P(b) = 0.3 × 0.5= 0.15

5.(b) ${}^4C_3\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^1=\frac{1}{4}$

6.(a) Leap year consist 52 weeks and 2 odd days.

So required probability = $\frac{{}^2C_2}{7}=\frac{1}{7}$

7.(a) Req. probability $=\frac{6}{36}=\frac{1}{6}$

8.(c) Let S be the sample space. Then,
n(S) = Number of ways of drawing 2 balls out of (7+3)

$= {}^{10}C_2 = \frac{(10\times 9)}{(2\times 1)} = 45$

Let E = Event of getting both balls of the same color. Then,
n (E) = Number of ways of drawing (2 balls out of 7) or (2 balls out of 3)

$= ({}^7C_2 + {}^3C_2) = (21 + 3) = 24.$

$\therefore$ P(E) $=\frac{n(E)}{n(S)}=\frac{24}{45}=\frac{8}{15}$

9. (c) Favourable cases = [(HHT), (HTT), (TTT), (TTH), (HTH), (THT), (THH)] Total no. of cases = 2^3 = 8
So probability = 7/8

10.(c) $P(A)P(\overline{B})+P(\overline{A})+P(B)=\frac{4}{5}\times\left(1-\frac{7}{10}\right)+\left(1-\frac{4}{5}\right)\times\frac{7}{10}:$

$=\frac{4}{5}\times\frac{3}{10}+\frac{1}{5}\times\frac{7}{10}=19/50$

11.(d) $n(S) = {}^{51}C_1 = 51$

Let A = the event that the drawn ticket contains a number which is a multiple of 2. and B = the event that the drawn ticket contains a number which is a multiple of 5.

$n(E)=n(A\cup B)=n(A)+n(B)-n(A\cap B)$

= 25 + 10 – 5 = 30

(where A $\cap$ B denotes the event that the drawn ticket contains a number which is a multiple of 2 and 5 both, i.e. 10)

$\therefore$ Required probability = $\frac{n(E)}{n(S)}=\frac{30}{51}=\frac{10}{17}$

12.(d) Required probability

$=\frac{{}^4C_1\times{}^9C_1}{{}^{13}C_2}+\frac{{}^4C_2}{{}^{13}C_2}=\frac{4\times 9}{78}+\frac{6}{78}=\frac{42}{78}=\frac{7}{13}$

13.(c) If he draws any combination of 3 socks he will definitely have the matching pair of either color.

14.(c) Required probability = $\frac{{}^5C_1\times{}^3C_1\times{}^{10}C_1}{{}^{18}C_3}=\frac{150}{816}=\frac{25}{136}$

15.(a) Required probability = P (a) + P (b) + P (c) (all the events are mutually exclusive)

$=\frac{1}{5}+\frac{1}{15}+\frac{1}{10}=\frac{11}{30}$

16.(c) Required probability

$=\frac{1}{4}+\frac{1}{5}+\frac{1}{6}+\frac{1}{7}=\frac{105+84+70+60}{420}=\frac{319}{420}$

17.(b) The question will be solved according to rule applied for selection without replacement.
It is given that first ball is found to be white hence white ball left is 12 and total ball left is 17. Hence required probability is 12/17

18.(c) P(A+B)= P(A)+P(B) - P(AB) = P(A)+ P(B)- P(A). P(B),
Because A and B are independent = 0.2 + 0.3 - 0.2 ×0.3 = 0.44

19.(d) Total pupils = 11, No. of boys = 5, So No. of girls = 11-5 = 6
As 2 girls are to be selected

Required probability = $\frac{{}^6C_2}{{}^{11}C_2}=\frac{6\times 5}{11\times 10}=\frac{3}{11}$

20.(a) P(A+B)= P(A)+P(B) For mutually exclusive events

Required probability = $\frac{2}{5}+\frac{1}{3}=\frac{11}{15}$

21.(d) n (S) = 6, n (a) = 3, hence P (a) = $\frac{3}{6}=\frac{1}{2}$

22.(c) $n(S) = {}^{10}C_2 = 45,\ n(A) = {}^{6}C_2 = 15$

$\therefore \quad P(A)=\frac{n(A)}{n(S)}=\frac{15}{45}=\frac{1}{3}$

23.(d) Required probability = $\frac{{}^{4}C_3\times{}^{6}C_2}{{}^{10}C_5}=\frac{60}{252}=\frac{5}{21}$

24.(b) The probability that first ticket shows the even number $=\frac{10}{21}$. Since the ticket is not replaced there are now 9 even numbered ticket and total 20 tickets left. Hence probability that second ticket shows the even number = $\frac{9}{20}$

Required probability = $\frac{10}{21}\times\frac{9}{20}=\frac{3}{14}$

25.(b) A = {2, 3, 5, 7, 11, 13, 17, 19, 23. 29}
$\therefore$ n(a) = 10, n (S) = 30

The required probability $P(A)=\frac{n(A)}{n(S)}=\frac{10}{30}=\frac{1}{3}$

26.(c) Out of 52 cards, one card can be drawn in ${}^{52}C_1$ ways
Hence (S) = 52.
Now let A be the required event. There are 2 cards which are red and queen *i.e.* red queen then $n(A) = {}^{2}C_1 = 2$.So, required probability

$=\frac{2}{52}=\frac{1}{26}$

27.(b) Three letters can be chosen out of 10 letters in ${}^{10}C_3$ ways. More than one vowels can be chosen in the following ways 2 vowels and one consonant or 3 vowels.
If A be the required event, then

$n(A) = {}^{4}C_2\times{}^{6}C_1+{}^{4}C_3$

Hence, required probability = $\frac{{}^{4}C_2\times{}^{6}C_1+{}^{4}C_3}{{}^{10}C_3}=\frac{1}{3}$

28.(d) Required probability = $\frac{8}{12}\times\frac{7}{11}=\frac{14}{33}$

29.(c) $P(A)=\frac{1}{2},\ P(B)=\frac{1}{3},\ P(C)=\frac{1}{4}$

Required probability = $P(A)\times P(B)\times P(\overline{C})$

$+P(\overline{A})\,P(B)\,P(C)+P(\overline{B})\,P(A)\,P(C)+P(A)\,P(B)\,P(C)$

$\left(\frac{1}{2}\times\frac{1}{3}\times\frac{3}{4}\right)+\left(\frac{1}{2}\times\frac{1}{3}\times\frac{1}{4}\right)+\left(\frac{2}{3}\times\frac{1}{2}\times\frac{1}{4}\right)+\left(\frac{1}{2}\times\frac{1}{3}\times\frac{1}{4}\right)$:

$=\frac{1}{8}+\frac{1}{24}+\frac{1}{12}+\frac{1}{24}=\frac{7}{24}$

30.(d) Let A = Event that the boy.
B = Event that the friend

Then, P (a) = $\frac{1}{3}$ and P (b) = $\frac{1}{5}$

$\therefore\ P(\overline{A})=\left(1-\frac{1}{3}\right)=\frac{2}{3}$ and $P(\overline{B})=\left(1-\frac{1}{5}\right)=\frac{4}{5}$

$\therefore$ Required probability = P [(A and not B) or (B and not A)] = $P(A)\,P(\overline{B})+P(\overline{B})\,P(\overline{A})$

$=\left(\frac{1}{3}\times\frac{4}{5}\right)+\left(\frac{1}{5}\times\frac{2}{3}\right)=\frac{4}{15}+\frac{2}{15}=\frac{6}{15}=\frac{2}{5}$

31.(b) A: Number chosen is a perfect square.
B: Number is a multiple of 4.
A $\cap$ B: Number is a perfect square as well as a multiple of 4.
A $\cup$ B: Number is a multiple of 4 or a perfect square.

P (A) = $\frac{31}{1000}$ $\quad\therefore$ P (B) = $\frac{1}{4}$ $\Rightarrow$ P (A $\cap$ B) = $\frac{15}{1000}$

P (A$\cup$B) = P(A) + P(B) – P(A$\cap$B).

P (A$\cup$B) = $\frac{31}{1000}+\frac{1}{4}-\frac{15}{1000}=\frac{266}{1000}=\frac{133}{500}$.

Hence (b)

32.(c) Probability of getting plumbing contract = P (A) = $\frac{2}{3}$

Probability of not getting an electric contract

= P (B') = $\frac{5}{9}$

Hence, probability of getting an electric contract

= P(B) = 1 - $\frac{5}{9}=\frac{4}{9}$

Probability of getting at least one contract

= P (A$\cup$B) = $\frac{4}{5}$

P (A$\cap$B) = P (A) + P (B) – P (A$\cup$B)

P (A$\cap$ B) = $\frac{2}{3}+\frac{4}{9}-\frac{4}{5}=\frac{14}{45}$.

Hence (c).

33.(d) Out of 20 biased coins, 5 always show tails. So out of 10 unbiased coins, the event is at least (14 – 5) = 9 coins show up tails. Now sample space of this event $n(S) = 2^{10}$ [Since the outcome of 20 coins is already known].
Desired event E: (getting 9 tails or 10 tails)
$\therefore$ $n(E) = {}^{10}C_9 + {}^{10}C_{10} = 10 + 1 = 11$

$\therefore$ $P(E) = \frac{11}{2^{10}} = 11 \times 2^{-10}$.

Hence, (d)

34.(c) For the first card to be neither 5 nor 7, it should belong to one from the remaining cards i.e.,
52 – 4 – 4 = 44 cards.

$\therefore$ P (first card is neither 5 nor 7) = $\frac{44}{52} = \frac{11}{13}$

There are now 51 cards of which four are 5 and four are 7.

$\therefore$ P (second card in neither 5 nor 7) = $\frac{43}{51}$

$\therefore$ P (Neither of two is 5 or 7) = $\frac{11}{13} \times \frac{43}{51} = \frac{473}{663}$.

Hence (c)

35.(b) Obviously if there are three mutually exclusive events, then probability of each lies between 0 and 1 (both inclusive).

$\therefore$ $0 \le \frac{1-p}{3} \le 1$, $0 \le \frac{4p-1}{2} \le 1$ and $0 \le \frac{6p-1}{4} \le 1$

$\therefore$ $-2 \le p \le 1$, $\frac{1}{4} \le p \le \frac{3}{4}$ and $\frac{1}{6} \le p \le \frac{5}{6}$

Also, $0 \le \left(\frac{1-p}{3} + \frac{4p-1}{2} + \frac{6p-1}{4}\right) \le 1$

$\frac{5}{38} \le p \le \frac{17}{38}$

Taking the intersection, we find that $P \in \left[\frac{1}{4}, \frac{17}{38}\right]$.

Hence, (b).

36.(d)

	C1	C2	C3	C4
R1				
R2				

The first row can be filled with the four colours in 4! Ways.
Once the first row is filled, the second row can be filled in d (4) ways, where d (4) is the number of derangements of 4.

$d(4) = 4!\left[\frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right] = 12 - 4 + 1 = 9$

$\therefore$ Total number of ways of filling the grid = 9 × 4!
Hence, (d).

37.(d) The movie Raju watched twice can be selected in ${}^6C_1$ = 6 ways.
Out of the five shows of this movie, Raju can select any two in ${}^5C_2 = 5 \times 2$ ways.
The remaining three movies he can see in ${}^5C_3 \times 3! = 5 \times 4 \times 3$ ways.
Thus, Raju can see the movies in $(5 \times 4 \times 3)^2$ ways.

$60 \times \frac{5!}{2!} = 5 \times 4 \times 3 \times \frac{5 \times 4 \times 3 \times 2}{2} = (5 \times 4 \times 3)^2$

Hence, (d).

38.(d)

Name	Amelia	Betty	Cindy	Dorothy	Ely	Frida
Birth date	21	2	19	31	18	5
Remainder	6	2	4	1	3	5
Seat no.	1	2	3	4	5	6

Thus, five ladies are "wrongly seated".
Total number of derangements for 5 objects.

$= 5!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!}\right]$

$= \frac{5!}{2!} - \frac{5!}{3!} + \frac{5!}{4!} - \frac{5!}{5!} = 60 - 20 + 5 - 1 = 44$

But the ladies are already "wrongly seated" in one way. Hence, (d)

39.(a) In all there are 3 + 4+ 6 = 13 books; they can be arranged in 13 ! ways.
A is the event that no two books on Microbiology are together.
The other seven books can be arranged in 7! Ways.
The 6 books on Microbiology can be placed between them or on their sides in ${}^8P_6$ ways.

$\therefore P(A) = \frac{7! \times {}^8P_6}{13!} = \frac{7! \times 4 \times 7!}{13!}$

All the books on any one subject are to be together; the we can think of all the books of anyone subject as a single object and n these as 3 objects can be arranged in 3! Ways.
Further, within themselves they can be arranged in 3!4!6! ways.
So, the total number of arrangements favorable to B = 3! × 3! × 4!× 6!

$\therefore$ $P(B) = \frac{3! \times 3! \times 4! \times 6!}{13!}$

So, the required ratio is

$\frac{P(A)}{P(B)} = \frac{7! \times 4 \times 7!}{13!} \times \frac{13!}{3! \times 3! \times 4! \times 6!}$

$= \frac{7! \times 4 \times 7!}{3! \times 3! \times 4! \times 6!} = \frac{7 \times 4 \times 7 \times 6 \times 5}{6 \times 6}$

$= \frac{490}{3}$. Hence, (a)

40.(d) Squares 1 and 5 must be selected.
Exactly one out of 2 and 4 can be selected.
Suppose 2 is selected then the remaining 2 squares can be (6,9), (8,9) (6,8) or (3, 8) – 4 ways. So, 2 × 4 = 8 ways. Hence, (d)

41.(d) To form a cube whose faces are of only two colours, we will have to leave out one row completely i.e., the cube has to be made out of 2 adjacent row, which is not possible. Hence, (d).

42.(a) Let Amit win 'a' games and Bhargav win 'b' games,
Then, Amit wins $\frac{2^a-1}{2-1} = 2^a - 1$ points and Bhargav wins $2^b - 1$ points.
Then , $2^a - 1 + 2^b - 1 = 2^a + 2^b - 2 = 574$
$\therefore\ 2^a + 2^b = 576$
Substituting values for 'a' from the options,
1. a = 6
$2^b = 512 \Rightarrow b = 9$; not possible since a > b
2. a = 7
$2^b = 448 \Rightarrow$ b is not an integer.
3. a = 8
$2^b = 320 \Rightarrow$ b is not an integer.
4. a = 9
$2^b = 64 \Rightarrow b = 6$
Only a = 9 satisfies the conditions. Hence, (a)

43.(c) P (A dies within 40 yrs) = $\frac{7}{12}$
P (B dies within 40 yrs) = $\frac{3}{5}$
The two events are independent.
Therefore, P (A and B both die within 40 yrs)
$= \frac{7}{12} \times \frac{3}{5} = \frac{7}{20}$
P (at least one of A and B both die within 40 yrs)
$= 1 - \frac{7}{20} = \frac{13}{20}$. Hence, (c)

44.(b) Let there be 100 people.
$\therefore$ 60 are males and 40 are females. Also 5% of 60 i.e. 3 are boys and 10% of 40 i.e. 4 are girls.
Now total number of boys = 3, and total number of children = 7.
$\therefore$ Probability that a selected child is a boy = 3/7

45.(b) Let P (A) = x and P(B) = y

$\Rightarrow x + y - P(A \cap B) \geq \frac{2}{5}$

$\Rightarrow x + y \geq \frac{2}{5} + P(A \cap B) \geq \frac{2}{5} + \frac{1}{10}$

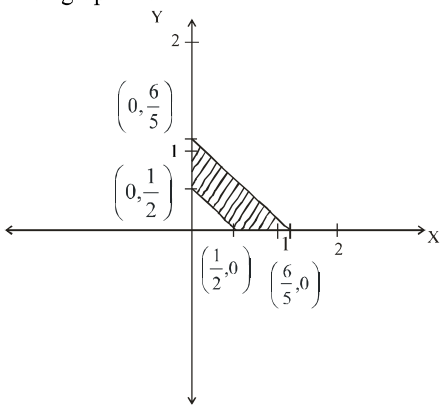
$\Rightarrow x + y \geq \frac{1}{2}\ P(A \cup B) \leq 1$

$\Rightarrow x + y - p(A \cap B) \leq 1$

$\Rightarrow x + y \leq 1 + P(A \cap B) = 1 + \frac{2}{10} = \frac{6}{5}$

$\Rightarrow x + y \leq \frac{6}{5}$

The graph is



$\Rightarrow$ Required Area = $\frac{1}{2} \times \frac{6}{5} \times \frac{6}{5} - \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$ a $= \frac{5}{8}$,
$= \frac{18}{25} - \frac{1}{8} = \frac{119}{200}$. Hence (b).

19 MENSURATION - 2D & 3D

Mensuration – 2D

1.(c) Let height = x cm then base = 2x cm
$2x \times x = 98 \Rightarrow 2x^2 = 98 \Rightarrow x = 7$ So height = 7 cm

2.(b) Given the side of the square be 2x. Then, its diagonal $= \sqrt{2} \times side = 2\sqrt{2}x$

Radius of in-circle = $\frac{2x}{2}$ = x, Area = 2πx

Radius of circum-circle = $\frac{2\sqrt{2}x}{2} = \sqrt{2}x$

$\frac{\text{Radius of in circle}}{\text{Radius of circum circle}} = \frac{x}{\sqrt{2}x} = \frac{1}{\sqrt{2}}$

3.(d) Length of diagonal = $\left(56 \times \frac{15}{60}\right)$m = 14 m

Sum of length and breadth = $\left(72 \times \frac{15}{60}\right)$m = 18 m

$\therefore\ \sqrt{l^2 + b^2} = 14\ or\ l^2 + b^2 = 196$ and $l + b = 18$.

Area = $lb = \frac{1}{2}(2lb) = \frac{1}{2}[(l+b)^2] - (l^2 + b^2)]$:

$= \frac{1}{2}[18]^2 - 196] = \frac{1}{2}(324 - 196) = 64\text{m}^2$